Problem Statement: Smart Campus Classroom Management System

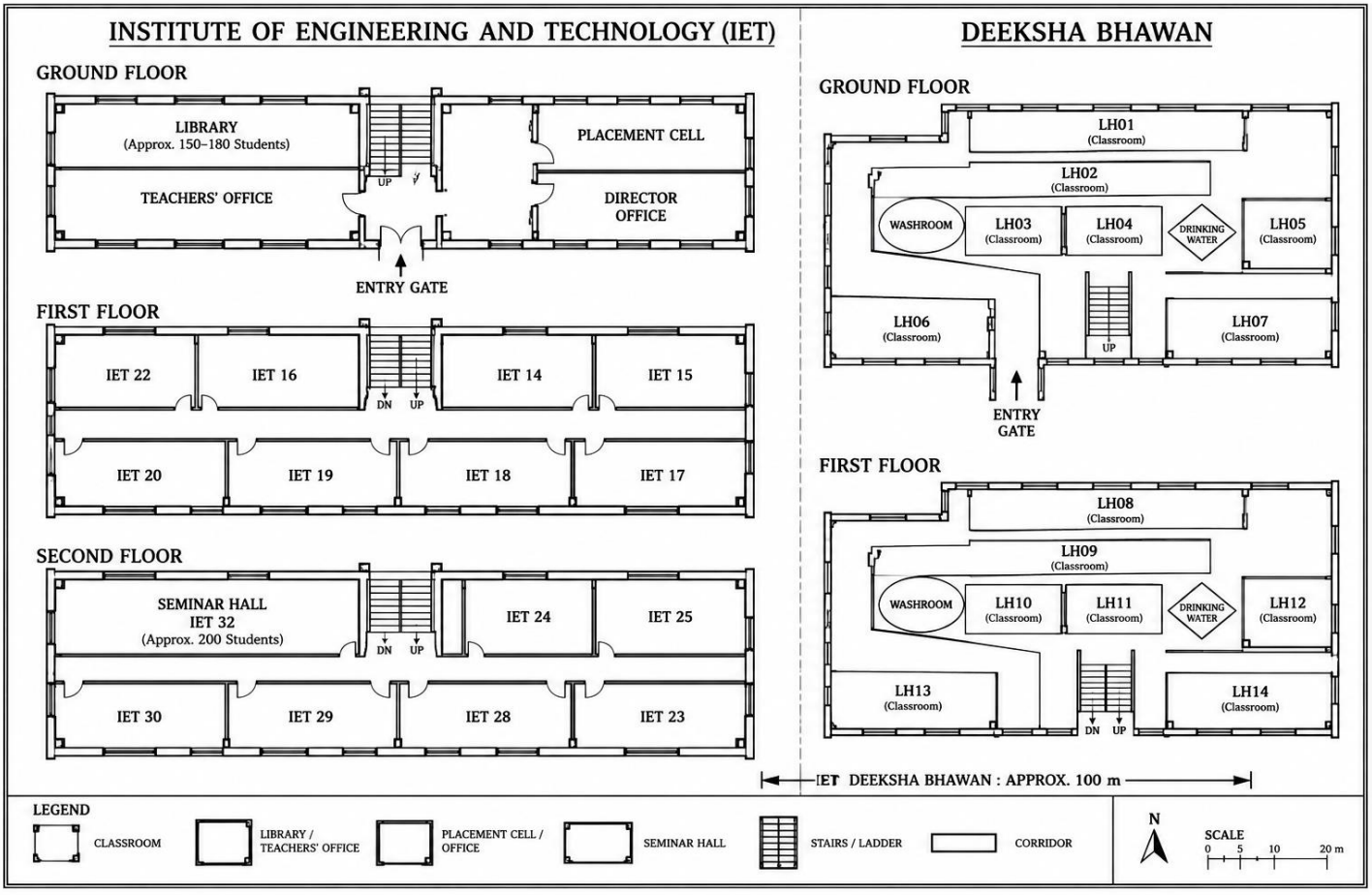
1. Problem Background

At **Deen Dayal Upadhyaya Gorakhpur University (DDUGU)**, the Institute of Engineering and Technology (IET) faces recurring challenges in classroom allocation and academic schedule coordination. Faculty members often struggle to find vacant classrooms for extra lectures, minor examinations, practical sessions, and events. Room bookings and schedule changes are not communicated through a single, systematic platform, causing confusion, overlapping allocations, and wasted time.

The consequences can be serious. **Last year, some students missed their minor examination (NSS) due to incorrect or insufficient information about the examination arrangements.** This highlights the need for a reliable system that coordinates classroom usage and ensures students

2. Campus Infrastructure and Supplied Dataset

Teams will receive student, branch, batch, and building datasets/blueprints. **Using the supplied datasets and blueprints of BOTH IET and Deeksha Bhawan in the prototype is mandatory.** The system must represent the actual campus infrastructure described below.



A. Institute of Engineering and Technology (IET)

IET accommodates students from six branches across the 1st, 2nd, 3rd, and 4th years:

- ✓ **CSE:** Approximately 150 students; divided into Sections A and B.
- ✓ **AI ML:** Approximately 150 students; divided into Sections A and B.
- ✓ **AI DS:** Approximately 75 students.
- ✓ **ECE:** Approximately 75 students.
- ✓ **ME:** Approximately 75 students.
- ✓ **IT:** Approximately 75 students.

Use the supplied student dataset to map students to their respective branches, academic years, batches, and sections.

Ground Floor

- Library: approximately 150–180 students; sometimes used for examinations.
- Faculty cabins, Director's Office, administrative office, and Placement Cell.

First Floor

- Four laboratories: CSE, ME, ECE, and IT.
- Four classrooms: **IET20, IET18, IET19, and IET17**.
- Each classroom has an approximate capacity of 75 students.

Second Floor

- Seminar Hall: **IET32**, with an approximate capacity of 200 students.
- Five classrooms: **IET24, IET25, IET29, IET30, and IET28**.

B. Deeksha Bhawan

Deeksha Bhawan is a separate building approximately 100 metres from IET. It has around 10–12 classrooms identified as **LH01–LH12** and is also used for examinations and other academic activities.

The platform must include Deeksha Bhawan as a separate building and allow faculty members to find suitable available rooms there when appropriate spaces in IET are occupied.

Room capacities, laboratory availability, and permitted uses should be based on supplied data or configurable records wherever exact information has not been provided.

3. Core Challenge

Develop a centralised, intelligent platform that enables faculty members to view, reserve, and manage classrooms across both buildings while giving students access to accurate academic schedules and announcements.

The solution should include:

1. **Room Availability:** Search by building, floor, room, date, time slot, duration, and required capacity.
2. **Faculty Booking:** Book rooms for extra classes, minor examinations, practical sessions, and events.

3. **Conflict Prevention:** Prevent overlapping bookings and suggest suitable alternative rooms.
4. **Schedule Updates:** Publish cancellations, rescheduling, examination notices, and other academic announcements.
5. **Student Access:** Let students view schedules and comment on or report conflicting or outdated information without modifying official bookings.
6. **Role-Based Access:** Provide appropriate permissions for faculty members, students, and administrators.
7. **Accountability:** Maintain a history of important bookings and changes to help resolve misunderstandings.
8. **Intelligent Assistance:** Use AI or scheduling algorithms to recommend suitable rooms and time slots, validating recommendations against actual booking records.

4. Expected Deliverable

Build a working prototype that demonstrates room discovery, booking, conflict prevention, alternative-room suggestions, and academic updates.

The prototype must integrate the **provided IET and Deeksha Bhawan building blueprints and student/branch/batch datasets**. Teams must submit a working demonstration, source-code repository, and setup instructions. Test bookings may be added where necessary, but they must be clearly identified as test data.

5. Technology Stack

Teams may choose appropriate technologies, including:

- **Frontend:** React, Next.js, HTML, CSS, JavaScript/TypeScript.
- **Backend:** Python (FastAPI/Django), Node.js, or equivalent.
- **Database:** PostgreSQL, MySQL, or MongoDB.
- **AI:** AI APIs/models for intelligent recommendations, natural-language search, or scheduling assistance.
- **Algorithms:** Room allocation, constraint checking, and scheduling optimisation.
- **DevOps:** Git/GitHub, deployment, testing, environment configuration, and CI/CD.
- **Advanced/Optional:** C++ for specialised optimisation logic or GPU simulation for a justified computational experiment.

These are suggested technologies, not mandatory tools. Prioritise a reliable, functional solution over unnecessary complexity.

Objective

Build a reliable campus coordination system that helps faculty members find the right classroom at the right time, prevents scheduling conflicts, and reduces the risk of students missing important academic activities because of incorrect or delayed information.

The goal is to solve a real problem at DDUGU using its actual infrastructure data.